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Université: Institut Universitaire des Sciences (IUS)

TD N° 8: Administration Systèmes Informatiques

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Supervision avec Prometheus et Grafana

Contexte

- La startup DevCloud héberge une plateforme web à fort trafic.

Suite à plusieurs incidents (surcharge CPU, mémoire saturée, indisponibilité du service web), l'équipe souhaite mettre en place une solution de supervision moderne.

Objectif

Mettre en place une solution complète de monitoring permettant de surveiller :

- CPU
- RAM
- Espace disque
- Trafic réseau
- Service web Nginx

À l'aide de :

- Prometheus
- Grafana

Architecture

[Serveur Linux]



|— Node Exporter (métriques système)

|— Nginx Exporter (métriques web)



[Prometheus] ---> collecte



[Grafana] ---> visualisation

1. Installation de Prometheus

sudo apt update

1.1 Créer l'utilisateur système :

sudo useradd --no-create-home --shell /bin/false prometheus

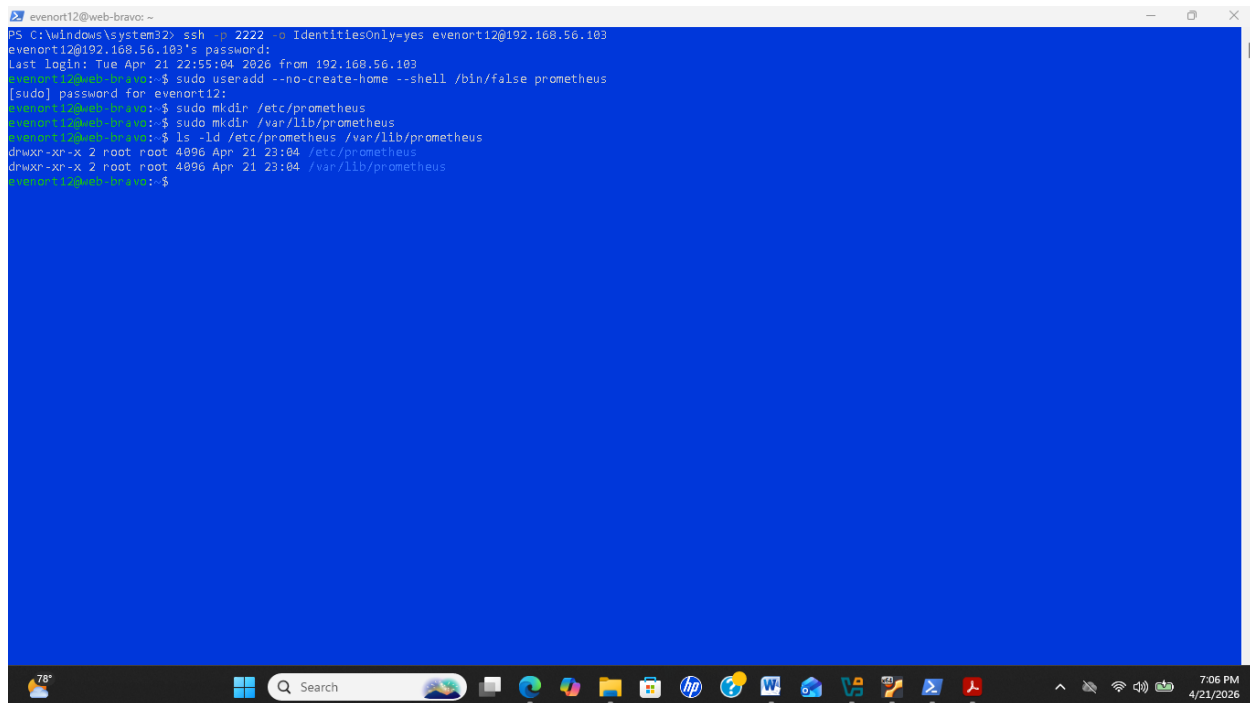
sudo chown -R prometheus:prometheus /etc/prometheus /var/lib/prometheus

1.2 Créer les dossiers :

sudo mkdir /etc/prometheus

sudo mkdir /var/lib/Prometheus

a) Vérification: `ls -ld /etc/prometheus /var/lib/prometheus`



```
PS C:\windows\system32> ssh -p 2222 -o IdentitiesOnly=yes evenort12@192.168.56.103
evenort12@192.168.56.103's password:
Last login: Tue Apr 21 22:55:04 2026 from 192.168.56.103
evenort12@web-bravo:~$ sudo useradd --no-create-home --shell /bin/false prometheus
[sudo] password for evenort12:
evenort12@web-bravo:~$ sudo mkdir /etc/prometheus
evenort12@web-bravo:~$ sudo mkdir /var/lib/prometheus
evenort12@web-bravo:~$ ls -ld /etc/prometheus /var/lib/prometheus
drwxr-xr-x 2 root root 4096 Apr 21 23:04 /etc/prometheus
drwxr-xr-x 2 root root 4096 Apr 21 23:04 /var/lib/prometheus
evenort12@web-bravo:~$
```

Figure 1: Cette image illustre l'étape de vérification des droits sur les répertoires de Prometheus.


```
evenort12@web-bravo:~$ sudo tar xvf prometheus-2.51.2.linux-amd64.tar.gz
prometheus-2.51.2.linux-amd64/
prometheus-2.51.2.linux-amd64/console_libraries/
prometheus-2.51.2.linux-amd64/console_libraries/menu.lib
prometheus-2.51.2.linux-amd64/console_libraries/prom.lib
prometheus-2.51.2.linux-amd64/promtool
prometheus-2.51.2.linux-amd64/NOTICE
prometheus-2.51.2.linux-amd64/LICENSE
prometheus-2.51.2.linux-amd64/consoles/
prometheus-2.51.2.linux-amd64/consoles/node-disk.html
prometheus-2.51.2.linux-amd64/consoles/index.html.example
prometheus-2.51.2.linux-amd64/consoles/node-overview.html
prometheus-2.51.2.linux-amd64/consoles/node.html
prometheus-2.51.2.linux-amd64/consoles/node-cpu.html
prometheus-2.51.2.linux-amd64/consoles/prometheus-overview.html
prometheus-2.51.2.linux-amd64/consoles/prometheus.html
prometheus-2.51.2.linux-amd64/prometheus.yml
prometheus-2.51.2.linux-amd64/prometheus
evenort12@web-bravo:~$ cd prometheus-2.51.2.linux-amd64.tar.gz
-bash: cd: prometheus-2.51.2.linux-amd64.tar.gz: Not a directory
evenort12@web-bravo:~$ ls -l
total 124252
drwxr-xr-x  4 prometheus      127    4096 avril 10  2024 prometheus-2.51.2.linux-amd64
-rw-rw-r--  1 evenort12 evenort12 101715261 avril 10  2024 prometheus-2.51.2.linux-amd64.tar.gz
-rw-rw-r--  1 evenort12 evenort12 25509454 avril 22  02:37 prometheus-2.51.2.linux-amd64.tar.gz.1
drwxrwxr-x  2 evenort12 evenort12    4096 avril 21  03:23 tp-consigne
evenort12@web-bravo:~$ sudo cp prometheus promtool /usr/local/bin/
cp: cannot stat 'prometheus': No such file or directory
cp: cannot stat 'promtool': No such file or directory
evenort12@web-bravo:~$ cd prometheus-2.51.2.linux-amd64
evenort12@web-bravo:~/prometheus-2.51.2.linux-amd64$ sudo cp prometheus promtool /usr/local/bin/
evenort12@web-bravo:~/prometheus-2.51.2.linux-amd64$ sudo cp prometheus.yml /etc/prometheus/
evenort12@web-bravo:~/prometheus-2.51.2.linux-amd64$ cp -r consoles/ console_libraries/ /etc/prometheus/
cp: cannot create directory '/etc/prometheus/consoles': Permission denied
cp: cannot create directory '/etc/prometheus/console_libraries': Permission denied
evenort12@web-bravo:~/prometheus-2.51.2.linux-amd64$ ls -l /usr/local/bin/prometheus /usr/local/bin/promtool
-rwxr-xr-x  1 root root 133616178 avril 22  03:56 /usr/local/bin/prometheus
-rwxr-xr-x  1 root root 124997437 avril 22  03:56 /usr/local/bin/promtool
evenort12@web-bravo:~/prometheus-2.51.2.linux-amd64$
```

```
Saving to: 'prometheus-2.51.2.linux-amd64.tar.gz'
prometheus-2.51.2.linux-amd64.tar.gz 100%[=====] 97,00M 595KB/s in 1m 41s
2026-04-22 03:45:19 (985 KB/s) - 'prometheus-2.51.2.linux-amd64.tar.gz' saved [101715261/101715261]
evenort12@web-bravo:~$ tar xvf prometheus-2.51.2.linux-amd64.tar.gz
prometheus-2.51.2.linux-amd64/
prometheus-2.51.2.linux-amd64/console_libraries/
prometheus-2.51.2.linux-amd64/console_libraries/menu.lib
tar: prometheus-2.51.2.linux-amd64/console_libraries/menu.lib: Cannot open: File exists
prometheus-2.51.2.linux-amd64/console_libraries/prom.lib
tar: prometheus-2.51.2.linux-amd64/console_libraries/prom.lib: Cannot open: File exists
prometheus-2.51.2.linux-amd64/promtool
tar: prometheus-2.51.2.linux-amd64/console_libraries: Cannot utime: Operation not permitted
tar: prometheus-2.51.2.linux-amd64/promtool: Cannot open: File exists
prometheus-2.51.2.linux-amd64/NOTICE
tar: prometheus-2.51.2.linux-amd64/NOTICE: Cannot open: Permission denied
prometheus-2.51.2.linux-amd64/LICENSE
tar: prometheus-2.51.2.linux-amd64/LICENSE: Cannot open: Permission denied
prometheus-2.51.2.linux-amd64/consoles/
tar: prometheus-2.51.2.linux-amd64/consoles: Cannot mkdir: Permission denied
prometheus-2.51.2.linux-amd64/consoles/node-disk.html
tar: prometheus-2.51.2.linux-amd64/consoles: Cannot mkdir: Permission denied
tar: prometheus-2.51.2.linux-amd64/consoles/node-disk.html: Cannot open: No such file or directory
prometheus-2.51.2.linux-amd64/consoles/index.html.example
tar: prometheus-2.51.2.linux-amd64/consoles: Cannot mkdir: Permission denied
tar: prometheus-2.51.2.linux-amd64/consoles/index.html.example: Cannot open: No such file or directory
prometheus-2.51.2.linux-amd64/consoles/node-overview.html
tar: prometheus-2.51.2.linux-amd64/consoles: Cannot mkdir: Permission denied
tar: prometheus-2.51.2.linux-amd64/consoles/node-overview.html: Cannot open: No such file or directory
prometheus-2.51.2.linux-amd64/consoles/node.html
tar: prometheus-2.51.2.linux-amd64/consoles: Cannot mkdir: Permission denied
tar: prometheus-2.51.2.linux-amd64/consoles/node.html: Cannot open: No such file or directory
prometheus-2.51.2.linux-amd64/consoles/node-cpu.html
tar: prometheus-2.51.2.linux-amd64/consoles: Cannot mkdir: Permission denied
tar: prometheus-2.51.2.linux-amd64/consoles/node-cpu.html: Cannot open: No such file or directory
prometheus-2.51.2.linux-amd64/consoles/prometheus-overview.html
tar: prometheus-2.51.2.linux-amd64/consoles: Cannot mkdir: Permission denied
tar: prometheus-2.51.2.linux-amd64/consoles/prometheus-overview.html: Cannot open: No such file or directory
```

figure 2.2.4 : Ces image montrent que le téléchargement de Prometheus a été réalisé (voir pages 5 et 6).

1.2 Copier les fichiers:

```
sudo cp prometheus /usr/local/bin/
```

```
sudo cp promtool /usr/local/bin/
```

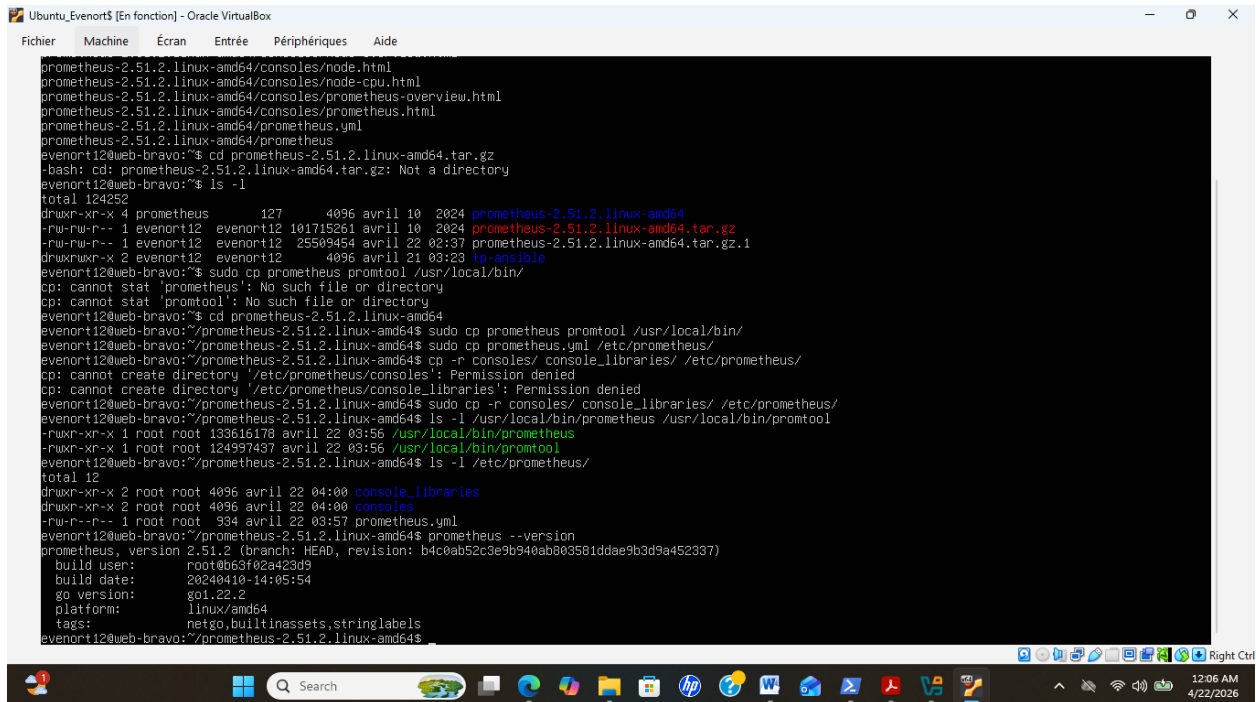
```
sudo cp prometheus.yml /etc/prometheus/
```

2. Test de vérification:

```
ls -l /usr/local/bin/prometheus /usr/local/bin/promtool
```

```
ls -l /etc/prometheus/(Tcheke konfigurasyon an)
```

```
prometheus --version
```



```
prometheus-2.51.2.linux-amd64/conssoles/node.html
prometheus-2.51.2.linux-amd64/conssoles/node-cpu.html
prometheus-2.51.2.linux-amd64/conssoles/prometheus-overview.html
prometheus-2.51.2.linux-amd64/conssoles/prometheus.html
prometheus-2.51.2.linux-amd64/prometheus.yml
prometheus-2.51.2.linux-amd64/prometheus
evenort12@web-bravo:~$ cd prometheus-2.51.2.linux-amd64.tar.gz
-bash: cd: prometheus-2.51.2.linux-amd64.tar.gz: Not a directory
evenort12@web-bravo:~$ ls -l
total 124252
drwxr-xr-x 4 prometheus 127 4096 avril 10 2024 prometheus-2.51.2.linux-amd64
-rw-rw-r-- 1 evenort12 evenort12 101715261 avril 10 2024 prometheus-2.51.2.linux-amd64.tar.gz
-rw-rw-r-- 1 evenort12 evenort12 25509454 avril 22 02:37 prometheus-2.51.2.linux-amd64.tar.gz.1
drwxr-xr-x 2 evenort12 evenort12 4096 avril 21 03:23 tp-ams101e
evenort12@web-bravo:~$ sudo cp prometheus promtool /usr/local/bin/
cp: cannot stat 'prometheus': No such file or directory
cp: cannot stat 'promtool': No such file or directory
evenort12@web-bravo:~$ cd prometheus-2.51.2.linux-amd64
evenort12@web-bravo:~/prometheus-2.51.2.linux-amd64$ sudo cp prometheus promtool /usr/local/bin/
evenort12@web-bravo:~/prometheus-2.51.2.linux-amd64$ sudo cp prometheus.yml /etc/prometheus/
evenort12@web-bravo:~/prometheus-2.51.2.linux-amd64$ cp -r conssoles/ console_libraries/ /etc/prometheus/
cp: cannot create directory '/etc/prometheus/conssoles': Permission denied
cp: cannot create directory '/etc/prometheus/console_libraries': Permission denied
evenort12@web-bravo:~/prometheus-2.51.2.linux-amd64$ sudo cp -r conssoles/ console_libraries/ /etc/prometheus/
evenort12@web-bravo:~/prometheus-2.51.2.linux-amd64$ ls -l /usr/local/bin/prometheus
-rwxr-xr-x 1 root root 130616178 avril 22 03:56 /usr/local/bin/prometheus
-rwxr-xr-x 1 root root 124997487 avril 22 03:56 /usr/local/bin/promtool
evenort12@web-bravo:~/prometheus-2.51.2.linux-amd64$ ls -l /etc/prometheus/
total 12
drwxr-xr-x 2 root root 4096 avril 22 04:00 console_libraries
drwxr-xr-x 2 root root 4096 avril 22 04:00 conssoles
-rw-r--r-- 1 root root 934 avril 22 03:57 prometheus.yml
evenort12@web-bravo:~/prometheus-2.51.2.linux-amd64$ prometheus --version
prometheus, version 2.51.2 (branch: HEAD, revision: b4c0ab52c3e9b940ab003581ddae9b3d9a452337)
 build user: root@b63f02a423d9
 build date: 20240410-14:05:54
 go version: go1.22.2
 platform: linux/amd64
 tags: netgo,builtinassets,stringlabels
evenort12@web-bravo:~/prometheus-2.51.2.linux-amd64$
```

Figure 5: Cette image montre la copie des fichiers de Prometheus, suivie de l'exécution du test de vérification

2.1 Configuration principale

```
sudo nano /etc/prometheus/prometheus.yml
```

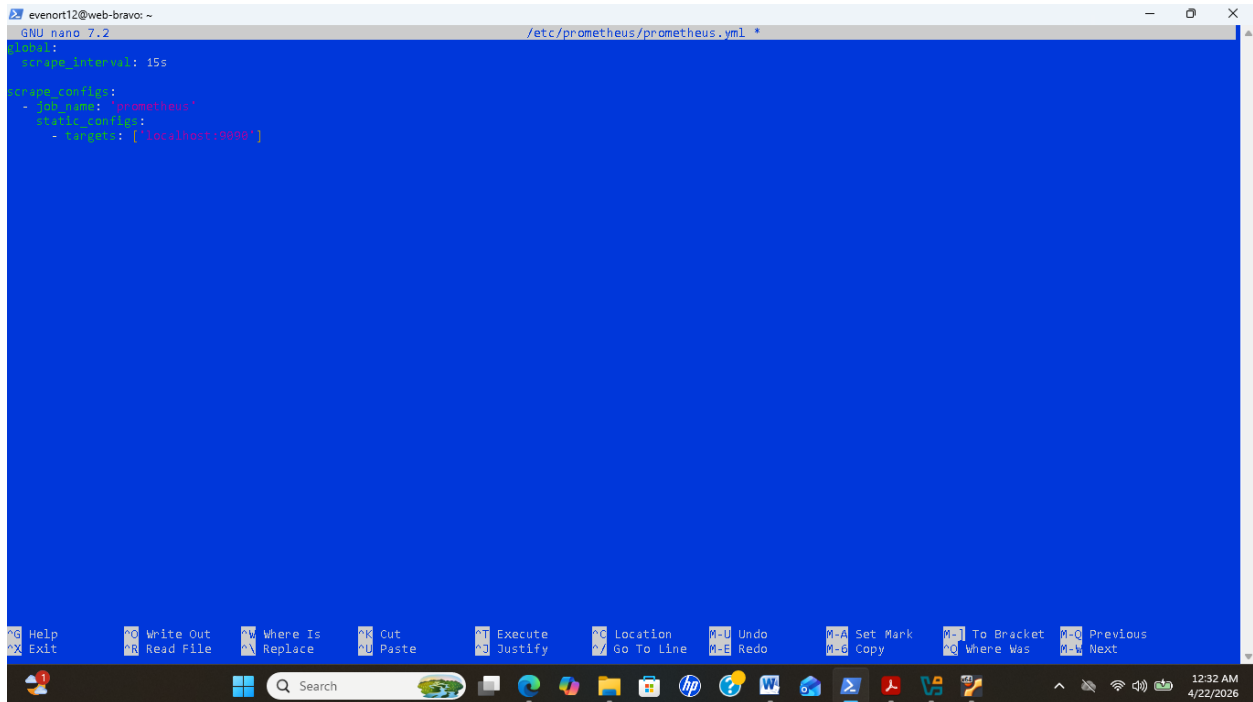
a) prometheus.yml

global:

scrape_interval: 15s

scrape_configs:

```
- job_name: 'prometheus'
static_configs:
  - targets: ['localhost:9090']
```



The screenshot shows a terminal window with the nano text editor open. The file being edited is `/etc/prometheus/prometheus.yml`. The configuration content is as follows:

```
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: 'prometheus'
    static_configs:
      - targets: ['localhost:9090']
```

The terminal window title is `evenort12@web-bravo: ~`. The bottom of the window shows the Windows taskbar with the time `12:32 AM 4/22/2026`.

Figure 6: Cette capture d'écran présente le fichier de configuration principal.

2.2 Service systemd

a) Créer : `sudo nano /etc/systemd/system/prometheus.service`

- Service

[Unit]

Description=Prometheus

After=network.target

[Service]

User=prometheus

ExecStart=/usr/local/bin/prometheus \

--config.file=/etc/prometheus/prometheus.yml \

--storage.tsdb.path=/var/lib/prometheus/

[Install]

WantedBy=multi-user.target


```
evenort12@web-bravo: ~  
GNU nano 7.2 /etc/systemd/system/prometheus.service *  
[Unit]  
Description=Prometheus  
Wants=network-online.target  
After=network-online.target  
  
[Service]  
User=prometheus  
Group=prometheus  
Type=simple  
ExecStart=/usr/local/bin/prometheus \\  
    --config.file=/etc/prometheus/prometheus.yml \\  
    --storage.tsdb.path=/var/lib/prometheus/ \\  
    --web.console.templates=/etc/prometheus/consoles \\  
    --web.console.libraries=/etc/prometheus/console_libraries  
  
[Install]  
WantedBy=multi-user.target
```

Figure 7: Cette image illustre la configuration du service Systemd.

2.3 Donner les permissions

```
sudo chown -R prometheus:prometheus /etc/prometheus  
sudo chown -R prometheus:prometheus /var/lib/prometheus  
sudo chmod -R 775 /var/lib/Prometheus
```

```
evenort12@web-bravo: ~  
PS C:\Windows\system32> ssh -i 2222 -o IdentitiesOnly=yes evenort12@192.168.56.103  
evenort12@192.168.56.103's password:  
Last login: Tue Apr 21 23:03:10 2026 from 192.168.56.1  
evenort12@web-bravo: ~  
evenort12@web-bravo:~$ sudo nano /etc/prometheus/prometheus.yml  
[sudo] password for evenort12:  
evenort12@web-bravo:~$ sudo useradd --no-create-home --shell /bin/false prometheus  
sudo chown -R prometheus:prometheus /etc/prometheus /var/lib/prometheus  
useradd: user 'prometheus' already exists  
evenort12@web-bravo:~$ sudo nano /etc/systemd/system/prometheus.service  
evenort12@web-bravo:~$ sudo chown -R prometheus:prometheus /etc/prometheus  
evenort12@web-bravo:~$ sudo chown -R prometheus:prometheus /var/lib/prometheus  
evenort12@web-bravo:~$ sudo chmod -R 775 /var/lib/prometheus  
evenort12@web-bravo:~$
```

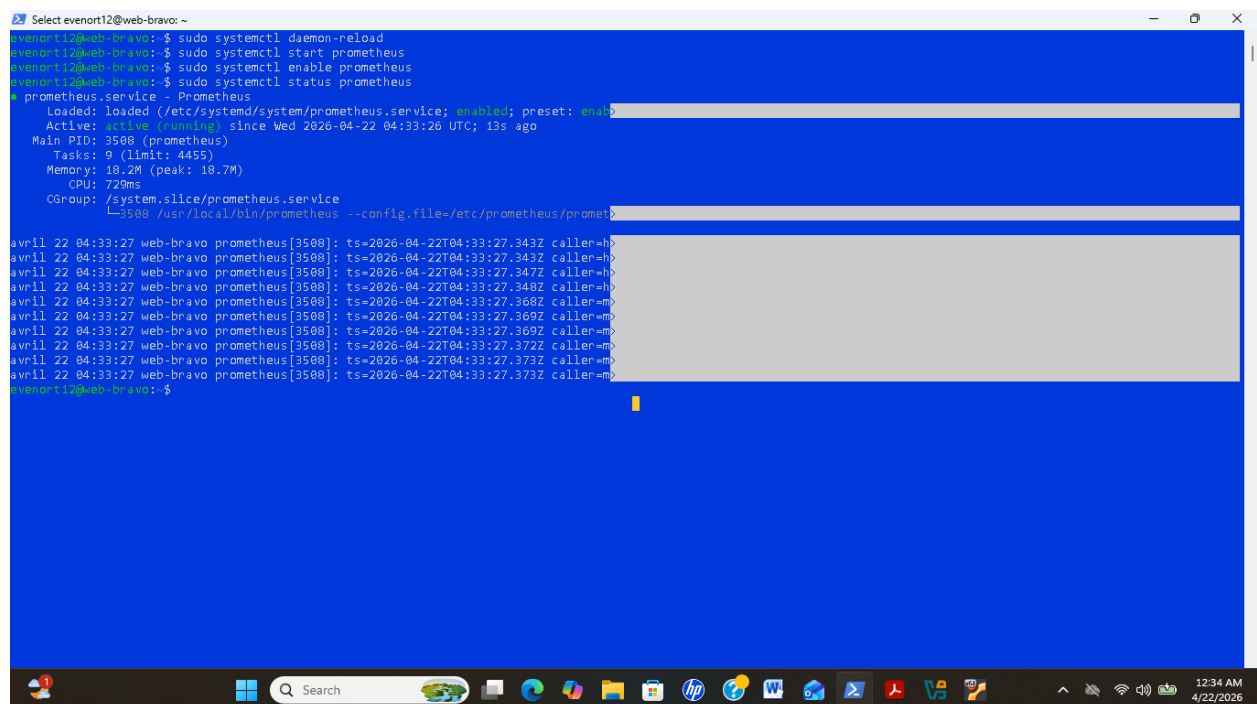
Figure 8 : Cette image montre l'attribution des permissions aux fichiers de Prometheus.

2.4 Démarrage

```
sudo systemctl daemon-reload
sudo systemctl start prometheus
sudo systemctl enable Prometheus
```

2.5 Vérification

```
sudo systemctl status Prometheus
```



```
Select evenort12@web-bravo: ~
evenort12@web-bravo:~$ sudo systemctl daemon-reload
evenort12@web-bravo:~$ sudo systemctl start prometheus
evenort12@web-bravo:~$ sudo systemctl enable prometheus
evenort12@web-bravo:~$ sudo systemctl status prometheus
● prometheus.service - Prometheus
   Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; preset: enab
   Active: active (running) since Wed 2026-04-22 04:33:26 UTC; 13s ago
     Main PID: 3508 (prometheus)
       Tasks: 9 (limit: 4455)
      Memory: 18.2M (peak: 18.7M)
         CPU: 720ms
    CGroup: /system.slice/prometheus.service
            └─3508 /usr/local/bin/prometheus --config.file=/etc/prometheus/promet

avril 22 04:33:27 web-bravo prometheus[3508]: ts=2026-04-22T04:33:27.343Z caller=h
avril 22 04:33:27 web-bravo prometheus[3508]: ts=2026-04-22T04:33:27.343Z caller=h
avril 22 04:33:27 web-bravo prometheus[3508]: ts=2026-04-22T04:33:27.347Z caller=h
avril 22 04:33:27 web-bravo prometheus[3508]: ts=2026-04-22T04:33:27.348Z caller=h
avril 22 04:33:27 web-bravo prometheus[3508]: ts=2026-04-22T04:33:27.368Z caller=m
avril 22 04:33:27 web-bravo prometheus[3508]: ts=2026-04-22T04:33:27.369Z caller=m
avril 22 04:33:27 web-bravo prometheus[3508]: ts=2026-04-22T04:33:27.369Z caller=m
avril 22 04:33:27 web-bravo prometheus[3508]: ts=2026-04-22T04:33:27.372Z caller=m
avril 22 04:33:27 web-bravo prometheus[3508]: ts=2026-04-22T04:33:27.373Z caller=m
avril 22 04:33:27 web-bravo prometheus[3508]: ts=2026-04-22T04:33:27.373Z caller=m
evenort12@web-bravo:~$
```

Figure 9: Cette capture d'écran confirme le bon fonctionnement du service Prometheus.

2.6 Interface Web

a) Accéder: <http://192.168.56.103:9090>

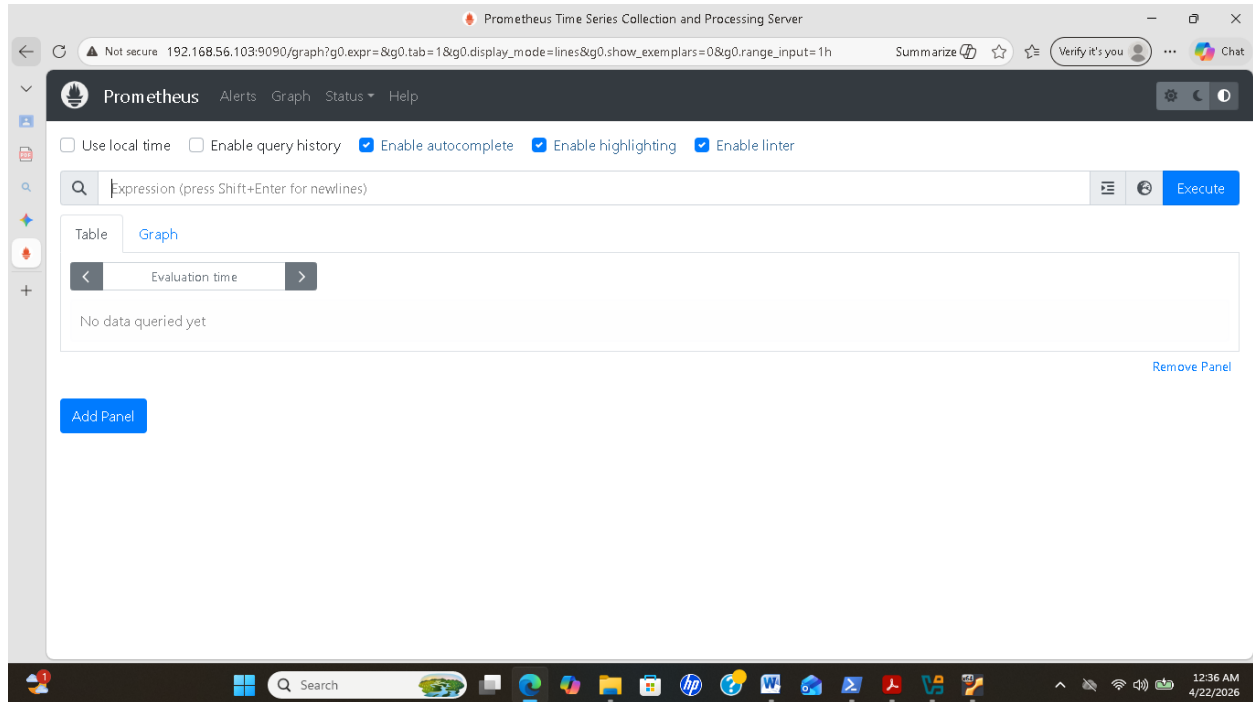


Figure 10 : J'ai accédé à l'interface web de Prometheus.

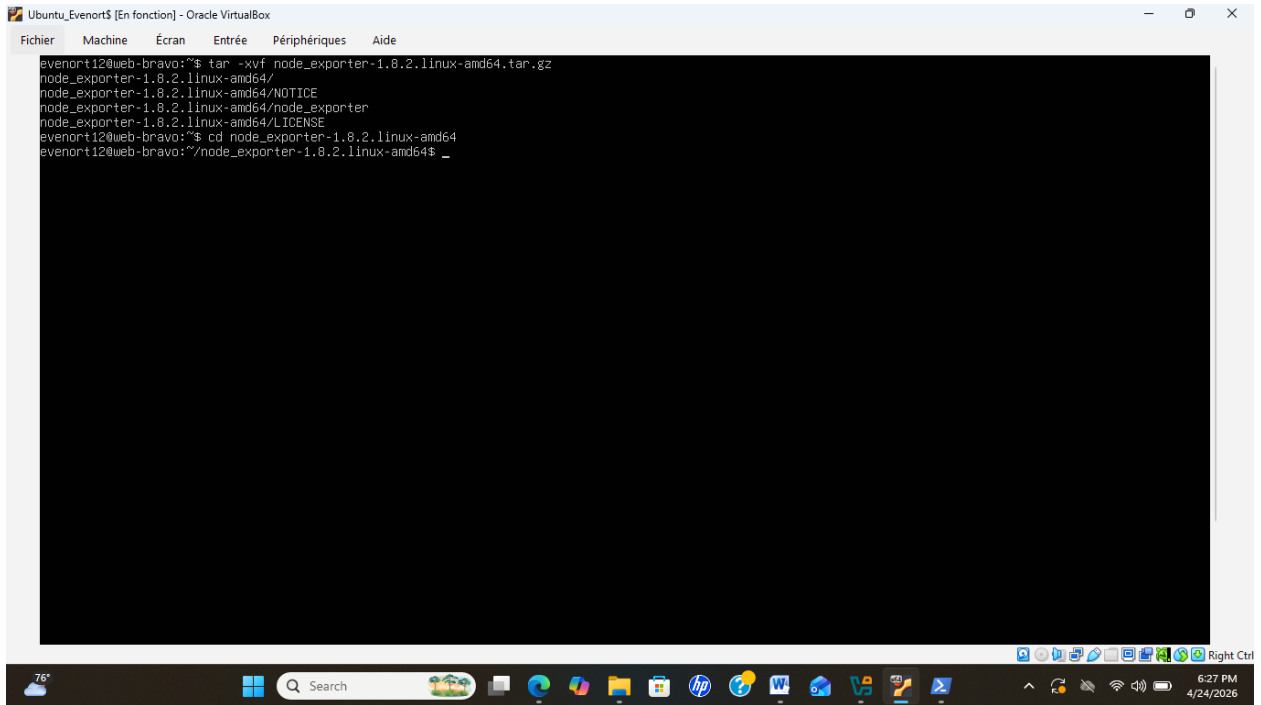


Figure 11, 12: ces images illustrent que Node Exporter a été installé correctement.

2. Service `node_exporter` : `sudo nano` `/etc/systemd/system/node_exporter.service`

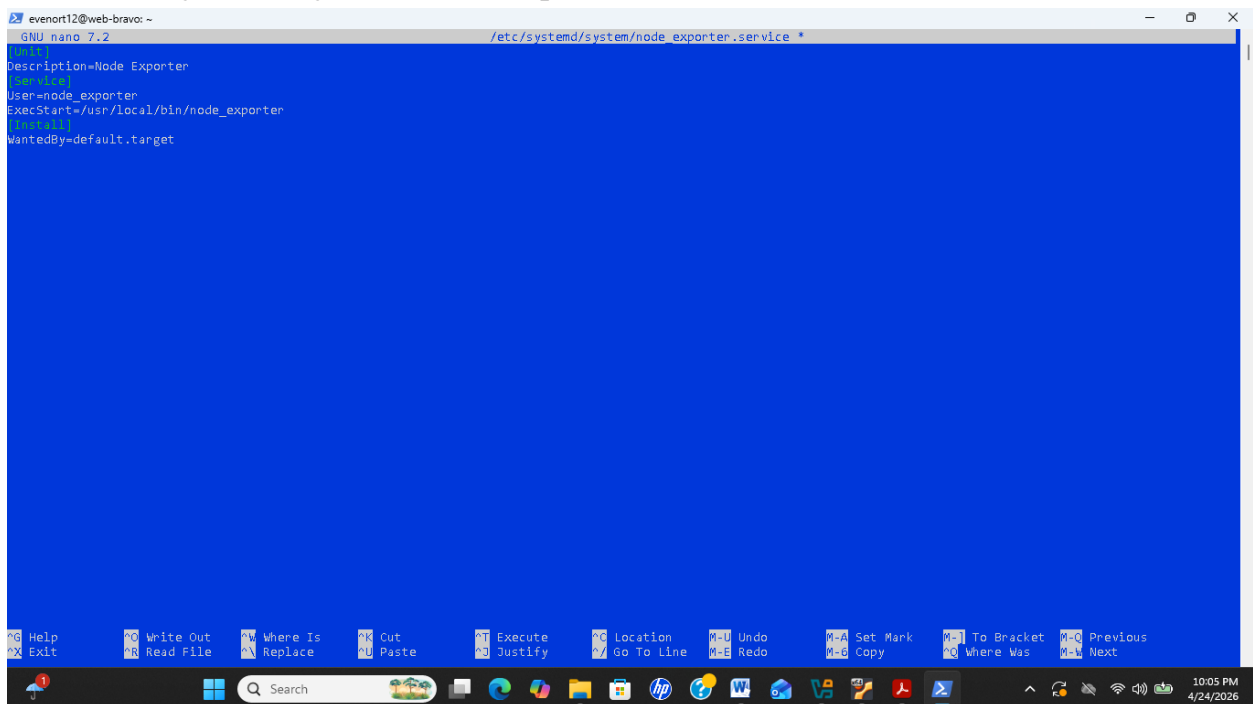
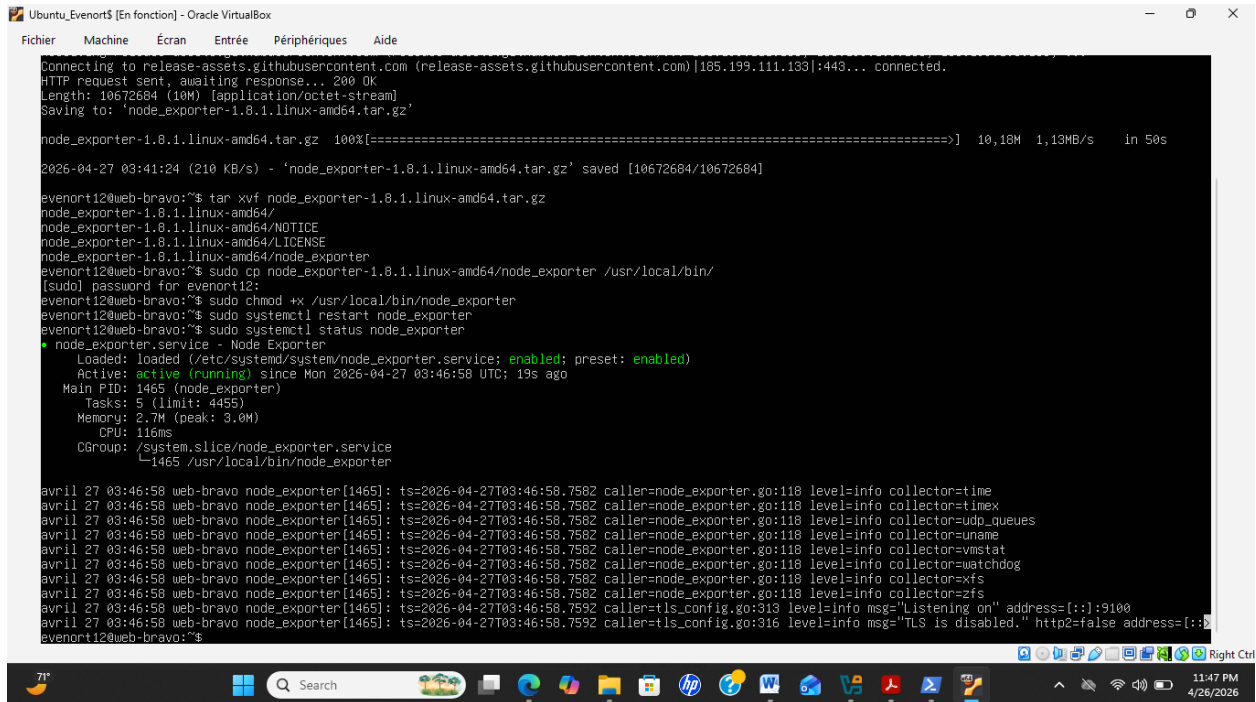


Figure 12 : Cette image atteste la bonne configuration du service `node_exporter`.

3. Démarrage

```
sudo systemctl daemon-reload
sudo systemctl start node_exporter
sudo systemctl enable node_exporter
sudo systemctl status node_exporter
```



```
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)[185.199.111.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10672684 (10M) [application/octet-stream]
Saving to: 'node_exporter-1.8.1.linux-amd64.tar.gz'

node_exporter-1.8.1.linux-amd64.tar.gz  100%[=====] 10,18M  1,13MB/s  in 50s

2026-04-27 03:41:24 (210 KB/s) - 'node_exporter-1.8.1.linux-amd64.tar.gz' saved [10672684/10672684]

evenort12@web-bravo:~$ tar xvf node_exporter-1.8.1.linux-amd64.tar.gz
node_exporter-1.8.1.linux-amd64/
node_exporter-1.8.1.linux-amd64/NOTICE
node_exporter-1.8.1.linux-amd64/LICENSE
node_exporter-1.8.1.linux-amd64/node_exporter
evenort12@web-bravo:~$ sudo cp node_exporter-1.8.1.linux-amd64/node_exporter /usr/local/bin/
[sudo] password for evenort12:
evenort12@web-bravo:~$ sudo chmod +x /usr/local/bin/node_exporter
evenort12@web-bravo:~$ sudo systemctl restart node_exporter
evenort12@web-bravo:~$ sudo systemctl status node_exporter
● node_exporter.service - Node Exporter
   Loaded: loaded (/etc/systemd/system/node_exporter.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-04-27 03:46:58 UTC; 19s ago
   Main PID: 1465 (node_exporter)
     Tasks: 5 (limit: 4455)
    Memory: 2.7M (peak: 3.0M)
       CPU: 116ms
   CGroup: /system.slice/node_exporter.service
           └─1465 /usr/local/bin/node_exporter

avril 27 03:46:58 web-bravo node_exporter[1465]: ts=2026-04-27T03:46:58.7582 caller=node_exporter.go:118 level=info collector=time
avril 27 03:46:58 web-bravo node_exporter[1465]: ts=2026-04-27T03:46:58.7582 caller=node_exporter.go:118 level=info collector=timex
avril 27 03:46:58 web-bravo node_exporter[1465]: ts=2026-04-27T03:46:58.7582 caller=node_exporter.go:118 level=info collector=udp_queues
avril 27 03:46:58 web-bravo node_exporter[1465]: ts=2026-04-27T03:46:58.7582 caller=node_exporter.go:118 level=info collector=uname
avril 27 03:46:58 web-bravo node_exporter[1465]: ts=2026-04-27T03:46:58.7582 caller=node_exporter.go:118 level=info collector=vmstat
avril 27 03:46:58 web-bravo node_exporter[1465]: ts=2026-04-27T03:46:58.7582 caller=node_exporter.go:118 level=info collector=watchdog
avril 27 03:46:58 web-bravo node_exporter[1465]: ts=2026-04-27T03:46:58.7582 caller=node_exporter.go:118 level=info collector=xfs
avril 27 03:46:58 web-bravo node_exporter[1465]: ts=2026-04-27T03:46:58.7582 caller=node_exporter.go:118 level=info collector=zfs
avril 27 03:46:58 web-bravo node_exporter[1465]: ts=2026-04-27T03:46:58.7592 caller=tls_config.go:313 level=info msg="Listening on" address=[::]:9100
avril 27 03:46:58 web-bravo node_exporter[1465]: ts=2026-04-27T03:46:58.7592 caller=tls_config.go:316 level=info msg="TLS is disabled." http2=false address=[::]:
evenort12@web-bravo:~$
```

Figure 13 : Redémarrage et vérification du statut du service node_exporter.

4. Ajouter Node Exporter à Prometheus

```
sudo nano /etc/prometheus/prometheus.yml
```

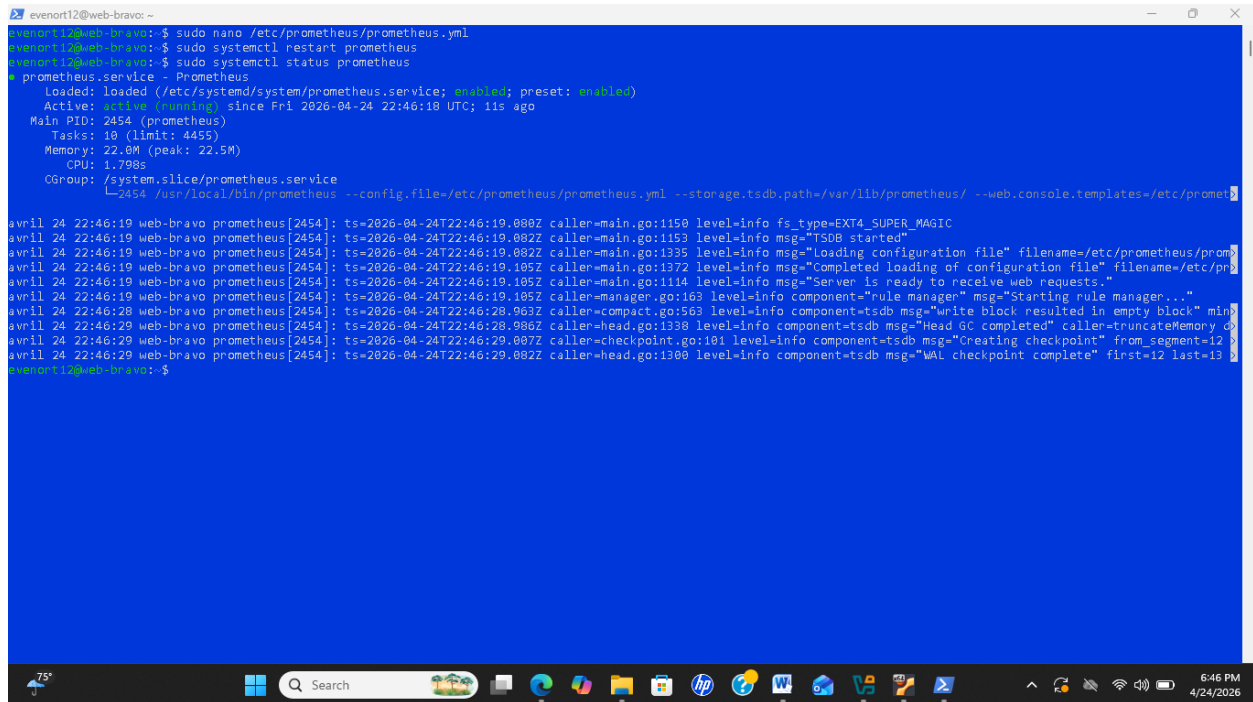
a) Ajout :

```
global:
  scrape_interval: 15s
  scrape_configs:
    - job_name: 'prometheus'
      static_configs:
        - targets: ['localhost:9090']
    - job_name: 'node_exporter'
      static_configs:
        - targets: ['localhost:9100']
```

b) Redémarrer

sudo systemctl restart prometheus

sudo systemctl status prometheus



```
evenort12@web-bravo: ~  
evenort12@web-bravo:~$ sudo nano /etc/prometheus/prometheus.yml  
evenort12@web-bravo:~$ sudo systemctl restart prometheus  
evenort12@web-bravo:~$ sudo systemctl status prometheus  
● prometheus.service - Prometheus  
   Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; preset: enabled)  
   Active: active (running) since Fri 2026-04-24 22:46:18 UTC; 11s ago  
     Main PID: 2454 (prometheus)  
       Tasks: 10 (limit: 4455)  
      Memory: 22.0M (peak: 22.5M)  
         CPU: 1.798s  
    CGroup: /system.slice/prometheus.service  
            └─2454 /usr/local/bin/prometheus --config.file=/etc/prometheus/prometheus.yml --storage.tsdb.path=/var/lib/prometheus/ --web.console.templates=/etc/prometheus/con  
avril 24 22:46:19 web-bravo prometheus[2454]: ts=2026-04-24T22:46:19.080Z caller=main.go:1150 level=info fs_type=EXT4_SUPER_MAGIC  
avril 24 22:46:19 web-bravo prometheus[2454]: ts=2026-04-24T22:46:19.082Z caller=main.go:1153 level=info msg="TSDB started"  
avril 24 22:46:19 web-bravo prometheus[2454]: ts=2026-04-24T22:46:19.082Z caller=main.go:1335 level=info msg="Loading configuration file" filename=/etc/prometheus/prometheus.yml  
avril 24 22:46:19 web-bravo prometheus[2454]: ts=2026-04-24T22:46:19.105Z caller=main.go:1372 level=info msg="Completed loading of configuration file" filename=/etc/prometheus/prometheus.yml  
avril 24 22:46:19 web-bravo prometheus[2454]: ts=2026-04-24T22:46:19.105Z caller=main.go:1114 level=info msg="Server is ready to receive web requests."  
avril 24 22:46:19 web-bravo prometheus[2454]: ts=2026-04-24T22:46:19.105Z caller=manager.go:163 level=info component="rule manager" msg="Starting rule manager..."  
avril 24 22:46:28 web-bravo prometheus[2454]: ts=2026-04-24T22:46:28.963Z caller=compact.go:563 level=info component=tsdb msg="write block resulted in empty block" min=0 max=0  
avril 24 22:46:29 web-bravo prometheus[2454]: ts=2026-04-24T22:46:28.986Z caller=head.go:1338 level=info component=tsdb msg="Head GC completed" caller=truncateMemory  
avril 24 22:46:29 web-bravo prometheus[2454]: ts=2026-04-24T22:46:29.007Z caller=checkpoint.go:101 level=info component=tsdb msg="Creating checkpoint" from_segment=12 to_segment=13  
avril 24 22:46:29 web-bravo prometheus[2454]: ts=2026-04-24T22:46:29.007Z caller=head.go:1300 level=info component=tsdb msg="WAL checkpoint complete" first=12 last=13  
evenort12@web-bravo:~$
```

Figure 14 : Cette image montre l'intégration de Node Exporter dans le fichier de configuration de Prometheus.

5. Vérification des targets

5.1 Accéder à l'interface Prometheus

<http://192.168.56.103:9090>

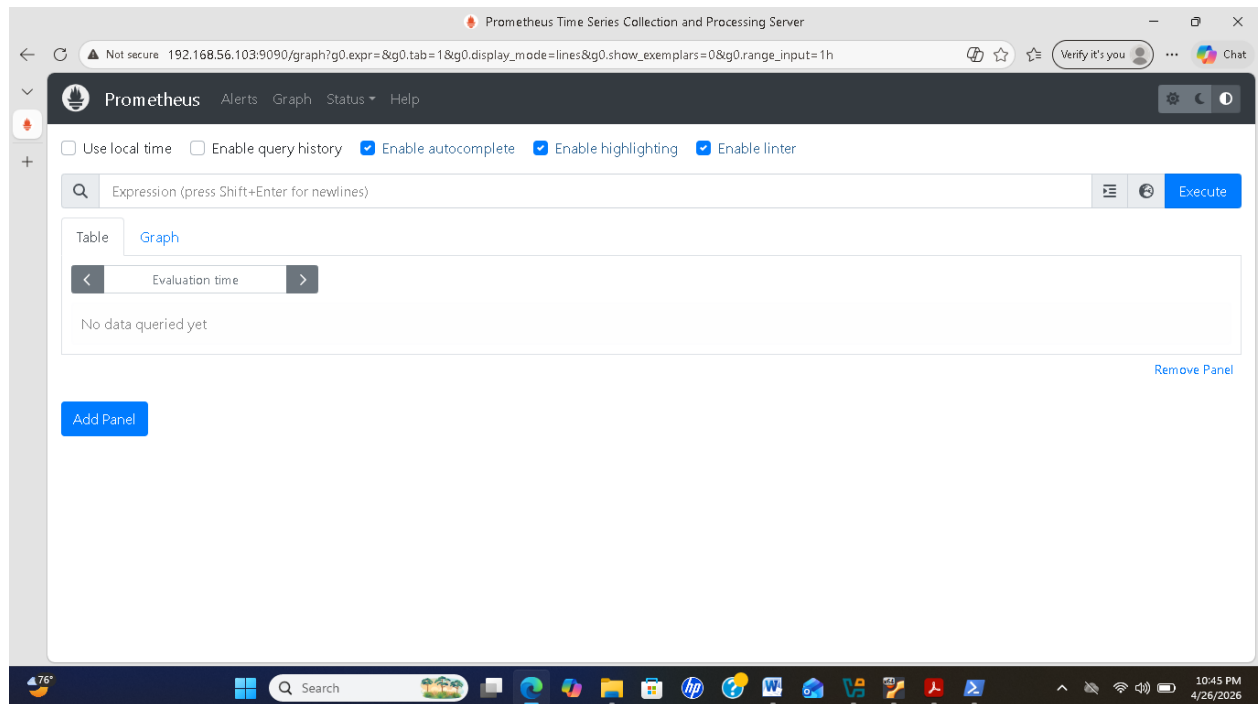


Figure 15: J'ai accédé à l'interface de Prometheus afin de vérifier les cibles (targets).

5.2 Vérifier l'état des services

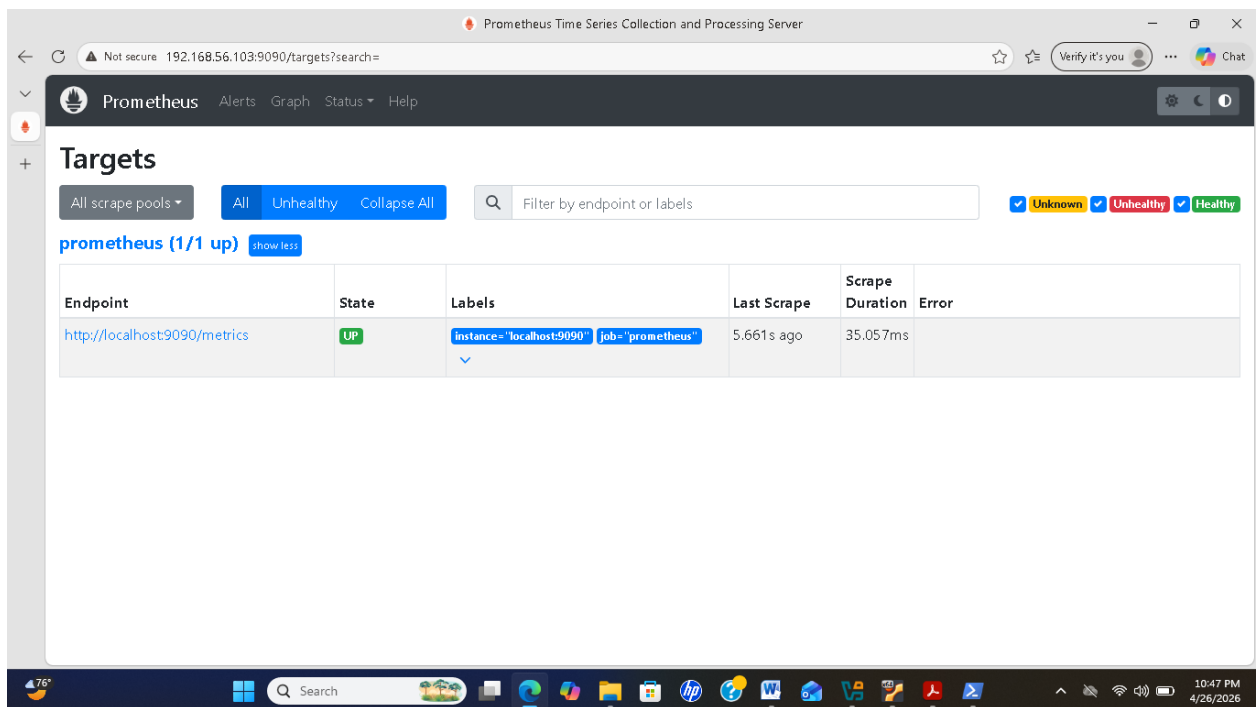
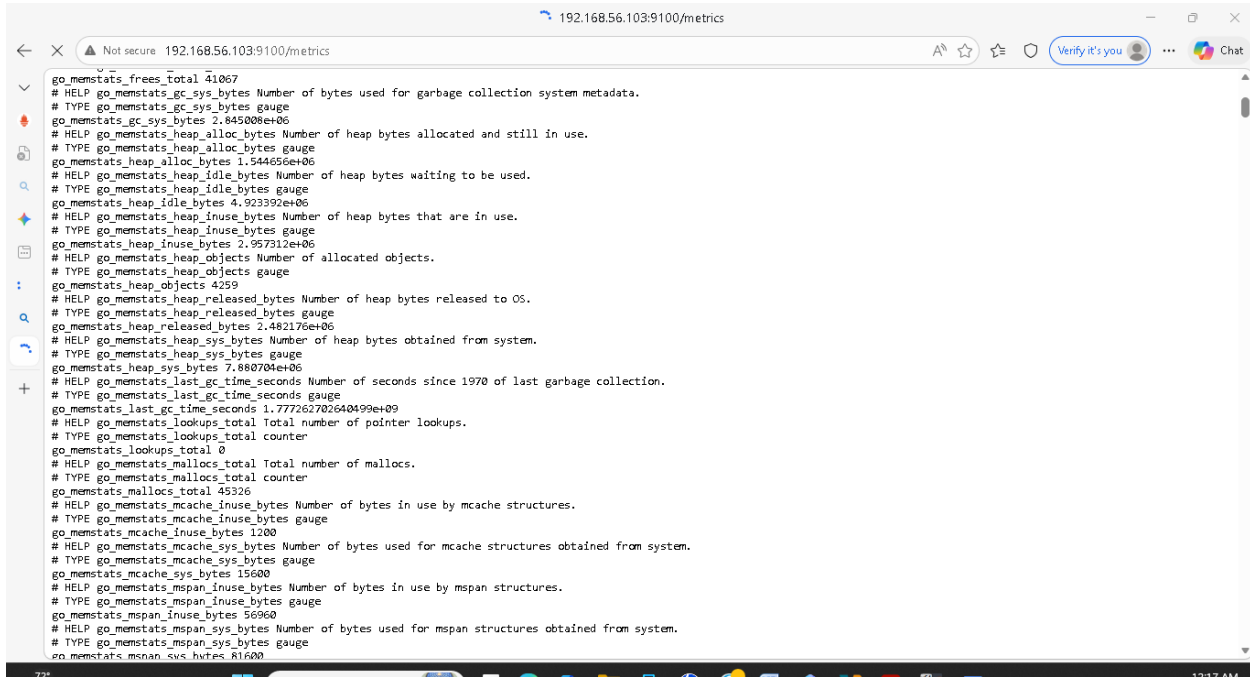


Figure 16 : Cette capture d'écran confirme le statut opérationnel des différents services.

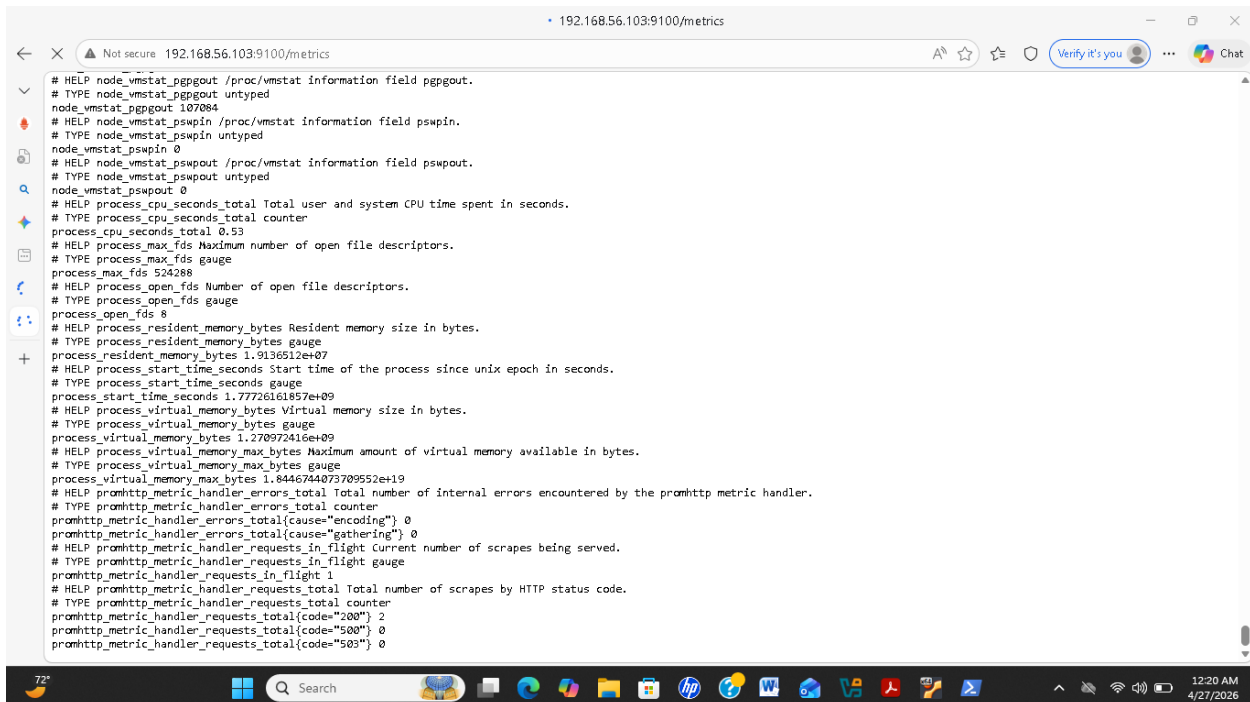
5.3 Tester manuellement les targets

Node Exporter

<http://localhost:9100/metrics>



```
go_memstats_frees_total 41067
# HELP go_memstats_gc_sys_bytes Number of bytes used for garbage collection system metadata.
# TYPE go_memstats_gc_sys_bytes gauge
go_memstats_gc_sys_bytes 2.845008e+06
# HELP go_memstats_heap_alloc_bytes Number of heap bytes allocated and still in use.
# TYPE go_memstats_heap_alloc_bytes gauge
go_memstats_heap_alloc_bytes 1.544656e+06
# HELP go_memstats_heap_idle_bytes Number of heap bytes waiting to be used.
# TYPE go_memstats_heap_idle_bytes gauge
go_memstats_heap_idle_bytes 4.923392e+06
# HELP go_memstats_heap_inuse_bytes Number of heap bytes that are in use.
# TYPE go_memstats_heap_inuse_bytes gauge
go_memstats_heap_inuse_bytes 2.957312e+06
# HELP go_memstats_heap_objects Number of allocated objects.
# TYPE go_memstats_heap_objects gauge
go_memstats_heap_objects 4259
# HELP go_memstats_heap_released_bytes Number of heap bytes released to OS.
# TYPE go_memstats_heap_released_bytes gauge
go_memstats_heap_released_bytes 2.482176e+06
# HELP go_memstats_heap_sys_bytes Number of heap bytes obtained from system.
# TYPE go_memstats_heap_sys_bytes gauge
go_memstats_heap_sys_bytes 7.880704e+06
# HELP go_memstats_last_gc_time_seconds Number of seconds since 1970 of last garbage collection.
# TYPE go_memstats_last_gc_time_seconds gauge
go_memstats_last_gc_time_seconds 1.777262702640499e+09
# HELP go_memstats_lookups_total Total number of pointer lookups.
# TYPE go_memstats_lookups_total counter
go_memstats_lookups_total 0
# HELP go_memstats_mallocs_total Total number of mallocs.
# TYPE go_memstats_mallocs_total counter
go_memstats_mallocs_total 45326
# HELP go_memstats_mcache_inuse_bytes Number of bytes in use by mcache structures.
# TYPE go_memstats_mcache_inuse_bytes gauge
go_memstats_mcache_inuse_bytes 1200
# HELP go_memstats_mcache_sys_bytes Number of bytes used for mcache structures obtained from system.
# TYPE go_memstats_mcache_sys_bytes gauge
go_memstats_mcache_sys_bytes 15600
# HELP go_memstats_mspan_inuse_bytes Number of bytes in use by mspan structures.
# TYPE go_memstats_mspan_inuse_bytes gauge
go_memstats_mspan_inuse_bytes 56960
# HELP go_memstats_mspan_sys_bytes Number of bytes used for mspan structures obtained from system.
# TYPE go_memstats_mspan_sys_bytes gauge
go_memstats_mspan_sys_bytes 81600
```



```
# HELP node_vmstat_ppgout /proc/vmstat information field ppgout.
# TYPE node_vmstat_ppgout untyped
node_vmstat_ppgout 107084
# HELP node_vmstat_pswpin /proc/vmstat information field pswpin.
# TYPE node_vmstat_pswpin untyped
node_vmstat_pswpin 0
# HELP node_vmstat_pswpout /proc/vmstat information field pswpout.
# TYPE node_vmstat_pswpout untyped
node_vmstat_pswpout 0
# HELP process_cpu_seconds_total Total user and system CPU time spent in seconds.
# TYPE process_cpu_seconds_total counter
process_cpu_seconds_total 0.53
# HELP process_max_fds Maximum number of open file descriptors.
# TYPE process_max_fds gauge
process_max_fds 524288
# HELP process_open_fds Number of open file descriptors.
# TYPE process_open_fds gauge
process_open_fds 8
# HELP process_resident_memory_bytes Resident memory size in bytes.
# TYPE process_resident_memory_bytes gauge
process_resident_memory_bytes 1.9136512e+07
# HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
# TYPE process_start_time_seconds gauge
process_start_time_seconds 1.77726161857e+09
# HELP process_virtual_memory_bytes Virtual memory size in bytes.
# TYPE process_virtual_memory_bytes gauge
process_virtual_memory_bytes 1.270972416e+09
# HELP process_virtual_memory_max_bytes Maximum amount of virtual memory available in bytes.
# TYPE process_virtual_memory_max_bytes gauge
process_virtual_memory_max_bytes 1.8446744873709552e+19
# HELP promhttp_metric_handler_errors_total Total number of internal errors encountered by the promhttp metric handler.
# TYPE promhttp_metric_handler_errors_total counter
promhttp_metric_handler_errors_total{cause="encoding"} 0
promhttp_metric_handler_errors_total{cause="gathering"} 0
# HELP promhttp_metric_handler_requests_in_flight Current number of scrapes being served.
# TYPE promhttp_metric_handler_requests_in_flight gauge
promhttp_metric_handler_requests_in_flight 1
# HELP promhttp_metric_handler_requests_total Total number of scrapes by HTTP status code.
# TYPE promhttp_metric_handler_requests_total counter
promhttp_metric_handler_requests_total{code="200"} 2
promhttp_metric_handler_requests_total{code="500"} 0
promhttp_metric_handler_requests_total{code="503"} 0
```

```
evenort12@web-bravo: ~  
GNU nano 7.2 /etc/prometheus/prometheus.yml *  
global:  
  scrape_interval: 15s  
  
scrape_configs:  
  - job_name: 'prometheus'  
    static_configs:  
      - targets: ['localhost:9090']  
  - job_name: 'mon-serveur-bravo'  
    static_configs:  
      - targets: ['192.168.56.103:9100']  
  
Help Write Out Where Is Cut Execute Location  
Exit Read File Replace Paste Justify Go To Line
```

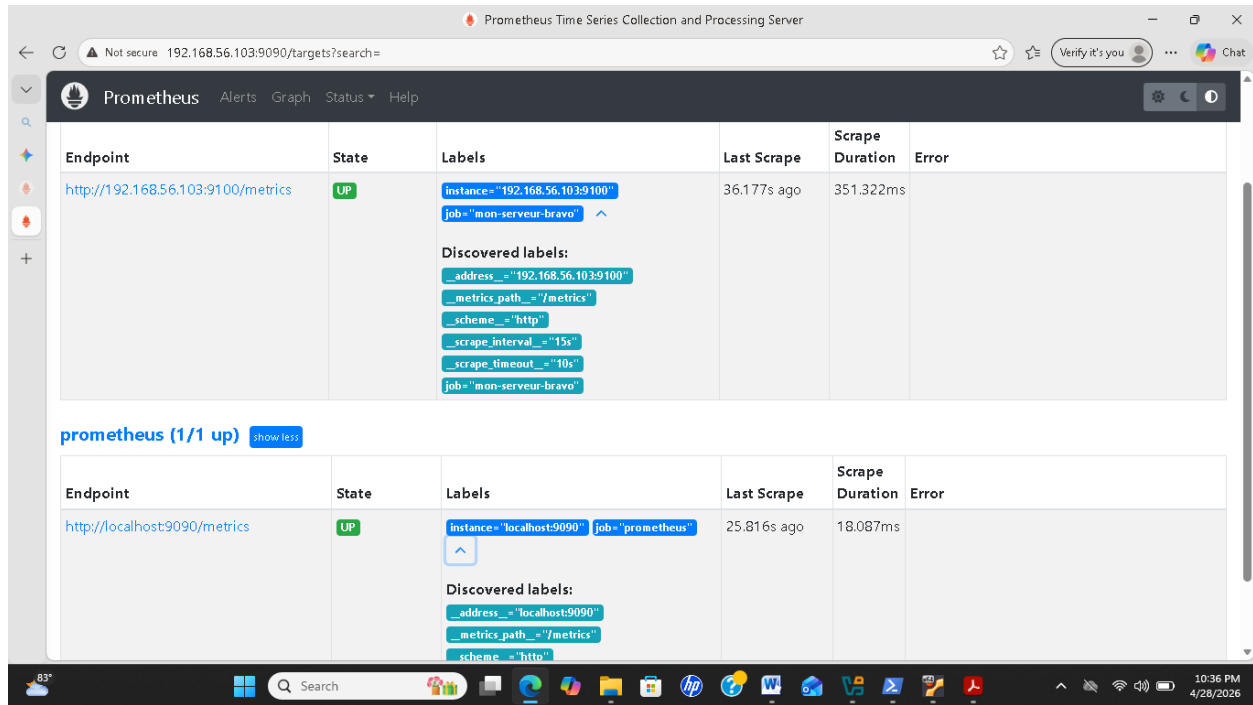
6. Redémarrer Prometheus

sudo systemctl restart Prometheus

```
evenort12@web-bravo: ~  
evenort12@web-bravo:~$ sudo systemctl restart prometheus  
[sudo] password for evenort12:  
evenort12@web-bravo:~$ sudo systemctl status prometheus  
● prometheus.service - Prometheus  
   Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; preset: enabled)  
   Active: active (running) since Wed 2026-04-29 02:51:32 UTC; 13s ago  
     Main PID: 3772 (prometheus)  
       Tasks: 10 (limit: 4455)  
      Memory: 28.5M (peak: 29.3M)  
         CPU: 2.819s  
    OGroup: /system.slice/prometheus.service  
           └─3772 /usr/local/bin/prometheus --config.file=/etc/prometheus/prometheus.yml --storage.tsdb.path=/var/lib/prometheus/ --web.console.templates=/etc/promet  
avril 29 02:51:34 web-bravo prometheus[3772]: ts=2026-04-29T02:51:34.191Z caller=main.go:1150 level=info fs_type=EXT4_SUPER_MAGIC  
avril 29 02:51:34 web-bravo prometheus[3772]: ts=2026-04-29T02:51:34.191Z caller=main.go:1153 level=info msg="TSDB started"  
avril 29 02:51:34 web-bravo prometheus[3772]: ts=2026-04-29T02:51:34.191Z caller=main.go:1335 level=info msg="Loading configuration file" filename=/etc/prometheus/prom  
avril 29 02:51:34 web-bravo prometheus[3772]: ts=2026-04-29T02:51:34.207Z caller=main.go:1372 level=info msg="Completed loading of configuration file" filename=/etc/prom  
avril 29 02:51:34 web-bravo prometheus[3772]: ts=2026-04-29T02:51:34.207Z caller=main.go:1114 level=info msg="Server is ready to receive web requests."  
avril 29 02:51:34 web-bravo prometheus[3772]: ts=2026-04-29T02:51:34.208Z caller=manager.go:163 level=info component="rule manager" msg="Starting rule manager..."  
avril 29 02:51:43 web-bravo prometheus[3772]: ts=2026-04-29T02:51:43.968Z caller=compact.go:563 level=info component=tsdb msg="write block resulted in empty block" min  
avril 29 02:51:43 web-bravo prometheus[3772]: ts=2026-04-29T02:51:43.977Z caller=head.go:1338 level=info component=tsdb msg="Head GC completed" caller=truncateMemory.d  
avril 29 02:51:43 web-bravo prometheus[3772]: ts=2026-04-29T02:51:43.987Z caller=checkpoint.go:181 level=info component=tsdb msg="Creating checkpoint" from_segment=28 >  
avril 29 02:51:44 web-bravo prometheus[3772]: ts=2026-04-29T02:51:44.077Z caller=head.go:1300 level=info component=tsdb msg="WAL checkpoint complete" first=20 last=29 >  
evenort12@web-bravo:~$
```

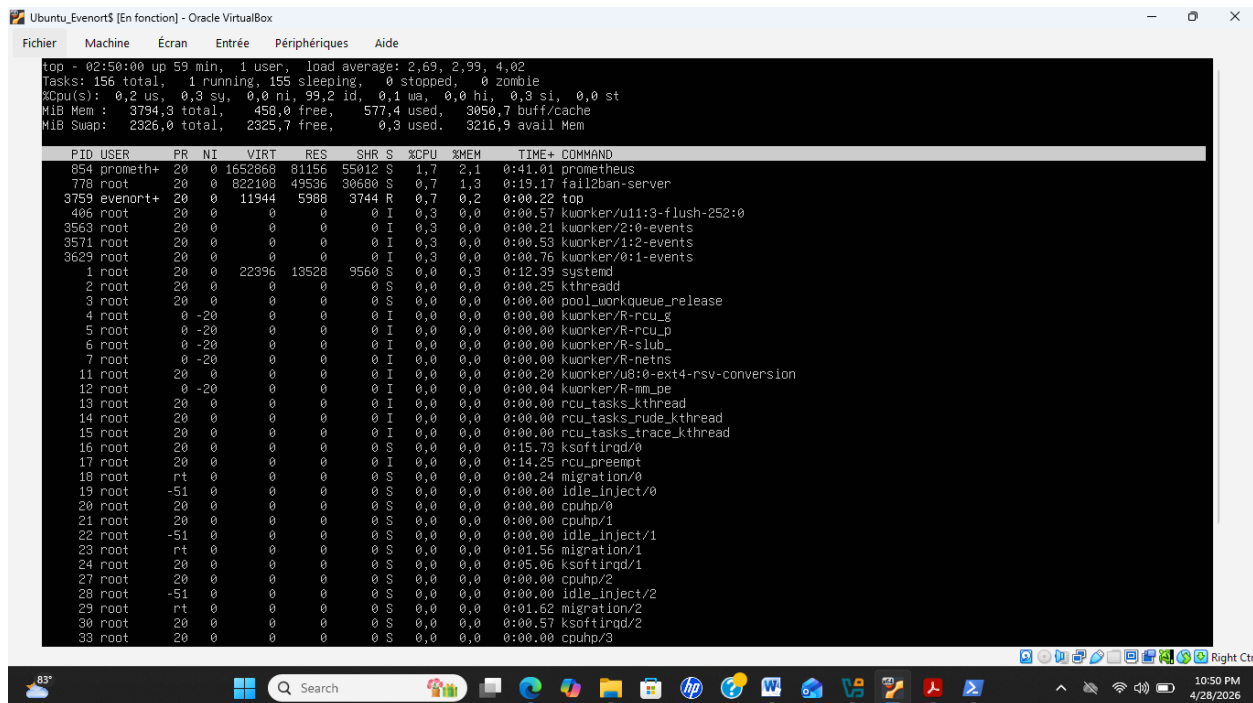
Figure 19 : Cette image montre le redémarrage du service Prometheus.

7. Vérification des cibles: <http://192.168.56.103:9090/targets>



8. Test de verification : top

8.1 arrêter: pkill yes



Installation de Grafana

```
sudo apt install -y software-properties-common apt-transport-https
```

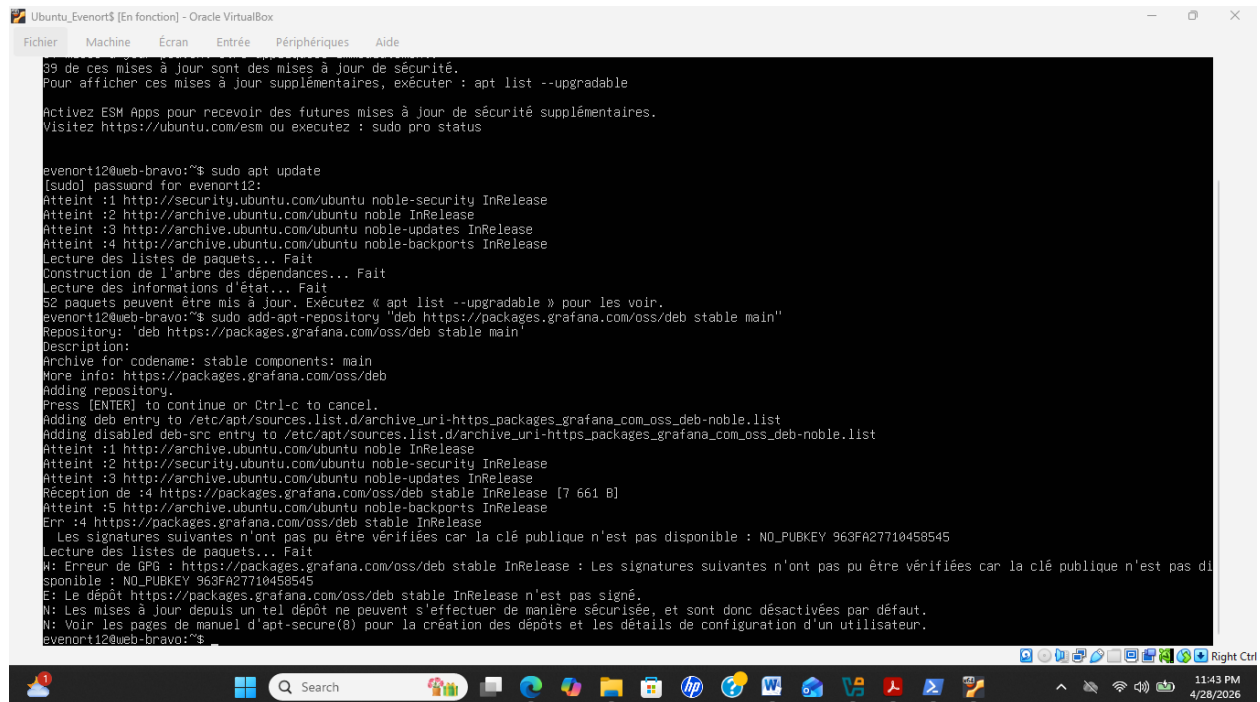
```
sudo add-apt-repository "deb https://packages.grafana.com/oss/deb stable main"
```

```
sudo apt update
```

```
sudo apt install grafana -y
```

```
sudo systemctl enable --now grafana-server
```

• J'ai Installé packages Grafana



```
Ubuntu_Evenort5 [En fonction] - Oracle VM VirtualBox
Fichier  Machine  Ecran  Entrée  Périphériques  Aide

39 de ces mises à jour sont des mises à jour de sécurité.
Pour afficher ces mises à jour supplémentaires, exécuter : apt list --upgradable

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou exécutez : sudo pro status

evenort12@web-bravo:~$ sudo apt update
[sudo] password for evenort12:
Atteint :1 http://security.ubuntu.com/ubuntu noble-security InRelease
Atteint :2 http://archive.ubuntu.com/ubuntu noble InRelease
Atteint :3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Atteint :4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
52 paquets peuvent être mis à jour. Exécutez « apt list --upgradable » pour les voir.
evenort12@web-bravo:~$ sudo add-apt-repository "deb https://packages.grafana.com/oss/deb stable main"
Repository: 'deb https://packages.grafana.com/oss/deb stable main'
Description:
Archive for codename: stable components: main
More info: https://packages.grafana.com/oss/deb
Adding repository.
Press [ENTER] to continue or Ctrl-C to cancel.
Adding deb entry to /etc/apt/sources.list.d/archive_uri-https_packages_grafana_com_oss_deb-noble.list
Adding disabled deb-src entry to /etc/apt/sources.list.d/archive_uri-https_packages_grafana_com_oss_deb-noble.list
Atteint :1 http://archive.ubuntu.com/ubuntu noble InRelease
Atteint :2 http://security.ubuntu.com/ubuntu noble-security InRelease
Atteint :3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Réception de :4 https://packages.grafana.com/oss/deb stable InRelease [7 661 B]
Atteint :5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Err :4 https://packages.grafana.com/oss/deb stable InRelease
  Les signatures suivantes n'ont pas pu être vérifiées car la clé publique n'est pas disponible : NO_PUBKEY 963FA27710458545
Lecture des listes de paquets... Fait
W: Erreur de GPG : https://packages.grafana.com/oss/deb stable InRelease : Les signatures suivantes n'ont pas pu être vérifiées car la clé publique n'est pas disponible : NO_PUBKEY 963FA27710458545
E: Le dépôt https://packages.grafana.com/oss/deb stable InRelease n'est pas signé.
N: Les mises à jour depuis un tel dépôt ne peuvent s'effectuer de manière sécurisée, et sont donc désactivées par défaut.
N: Voir les pages de manuel d'apt-secure(8) pour la création des dépôts et les détails de configuration d'un utilisateur.
evenort12@web-bravo:~$
```

Accès : <http://192.168.56.103:3000>

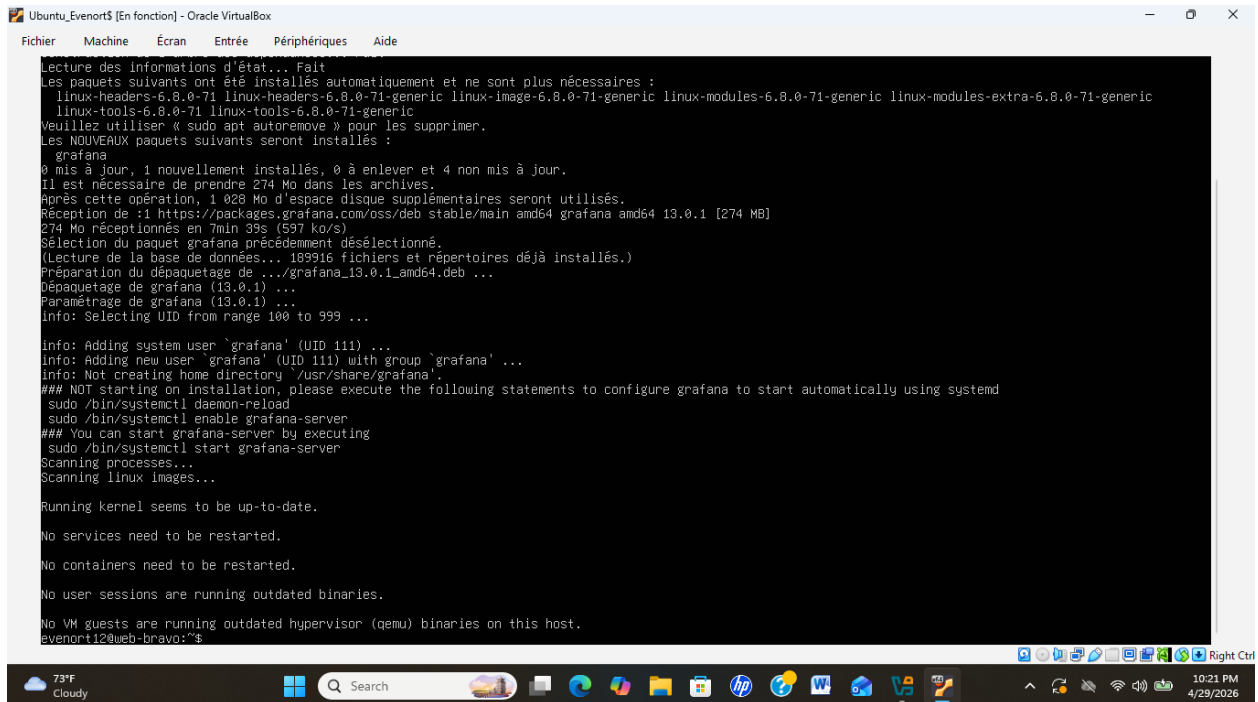
Login par défaut : admin / admin

3. J'ai Ajouté Prometheus comme source de données

Menu → Data Sources → Add Data Source → Prometheus

URL: <http://localhost:9090>

Save & Test → OK



```
Ubuntu_Evenort$ [En fonction] - Oracle VirtualBox
Fichier  Machine  Écran  Entrée  Périphériques  Aide

Lecture des informations d'état... Fait
Les paquets suivants ont été installés automatiquement et ne sont plus nécessaires :
  linux-headers-6.8.0-71 linux-headers-6.8.0-71-generic linux-image-6.8.0-71-generic linux-modules-6.8.0-71-generic linux-modules-extra-6.8.0-71-generic
linux-tools-6.8.0-71 linux-tools-6.8.0-71-generic
Veuillez utiliser « sudo apt autoremove » pour les supprimer.
Les NOUVEAUX paquets suivants seront installés :
  grafana
0 mis à jour, 1 nouvellement installés, 0 à enlever et 4 non mis à jour.
Il est nécessaire de prendre 274 Mo dans les archives.
Après cette opération, 1 028 Mo d'espace disque supplémentaires seront utilisés.
Réception de :1 https://packages.grafana.com/oss/deb/stable/main amd64 grafana amd64 13.0.1 [274 MB]
274 Mo réceptionnés en 7min 39s (597 ko/s)
Sélection du paquet grafana précédemment désélectionné.
(Lecture de la base de données... 108916 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../grafana_13.0.1_amd64.deb ...
Dépaquetage de grafana (13.0.1) ...
Paramétrage de grafana (13.0.1) ...
info: Selecting UID from range 100 to 999 ...

info: Adding system user `grafana' (UID 111) ...
info: Adding new user `grafana' (UID 111) with group `grafana' ...
info: Not creating home directory `/usr/share/grafana'.
### NOT starting on installation, please execute the following statements to configure grafana to start automatically using systemd
  sudo /bin/systemctl daemon-reload
  sudo /bin/systemctl enable grafana-server
### You can start grafana-server by executing
  sudo /bin/systemctl start grafana-server
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

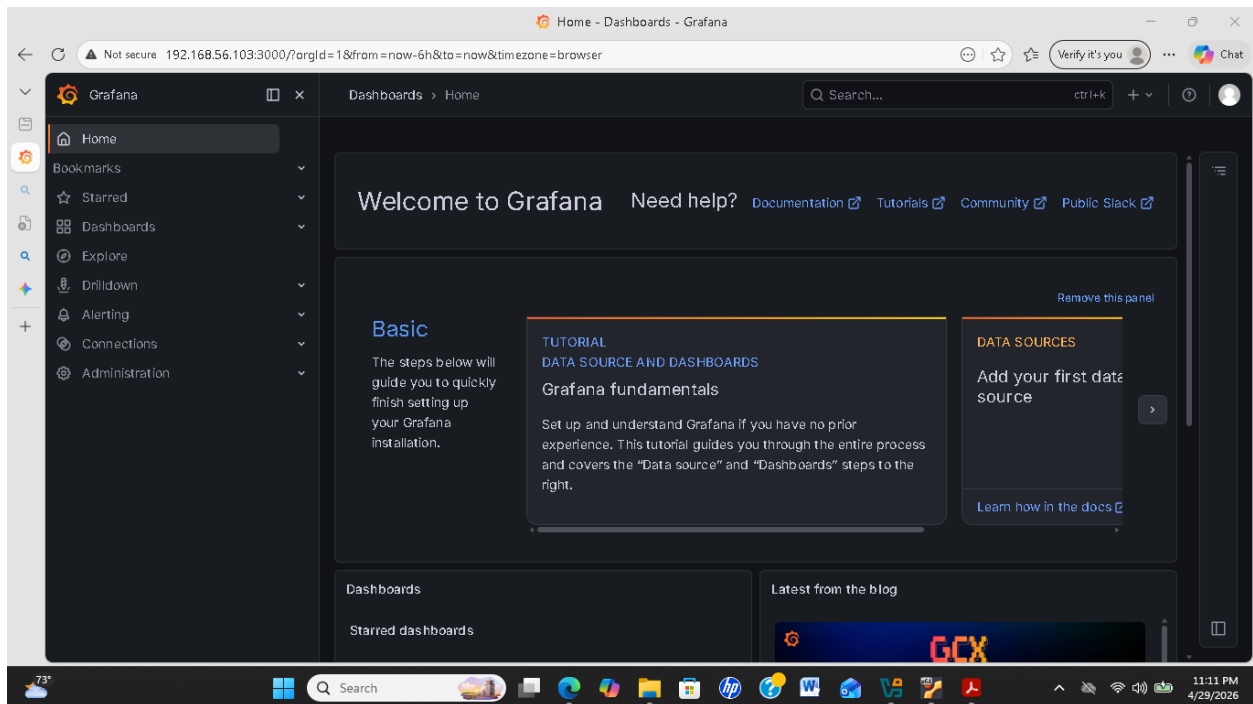
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
evenort12@web-bravo:~$
```

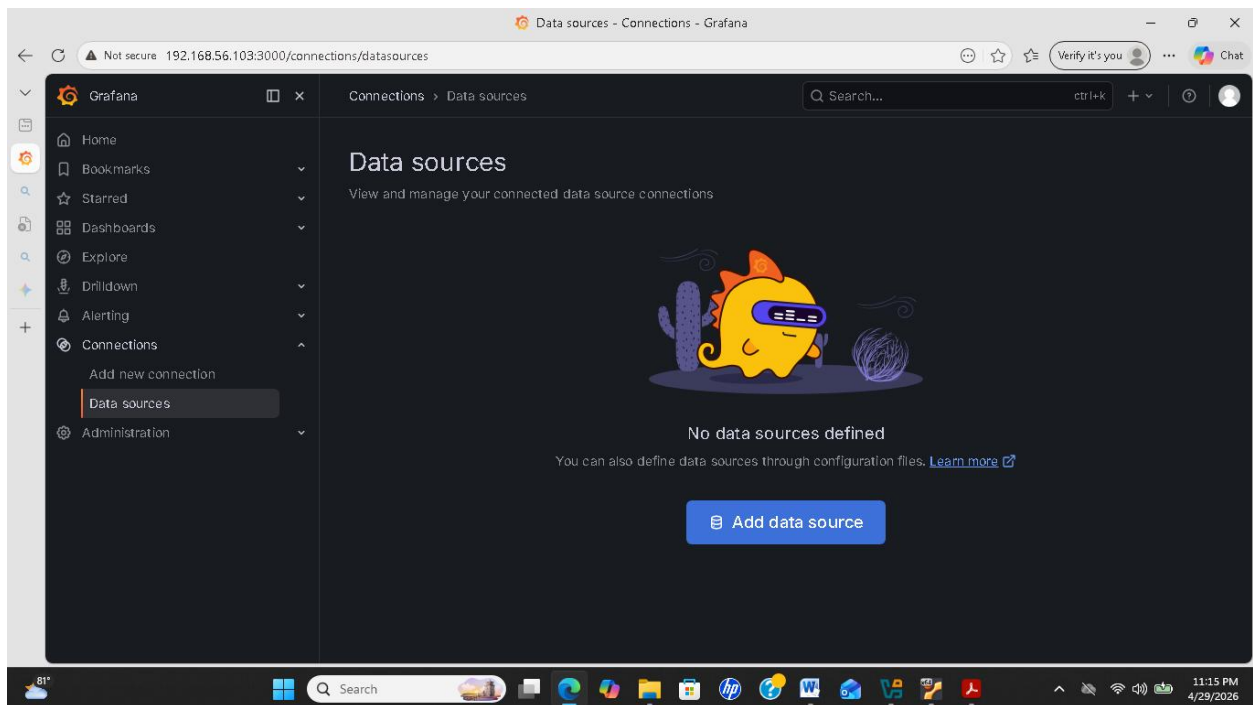
3 J'ai démarré Grafana et vérifiant fonctionnement du service Grafana

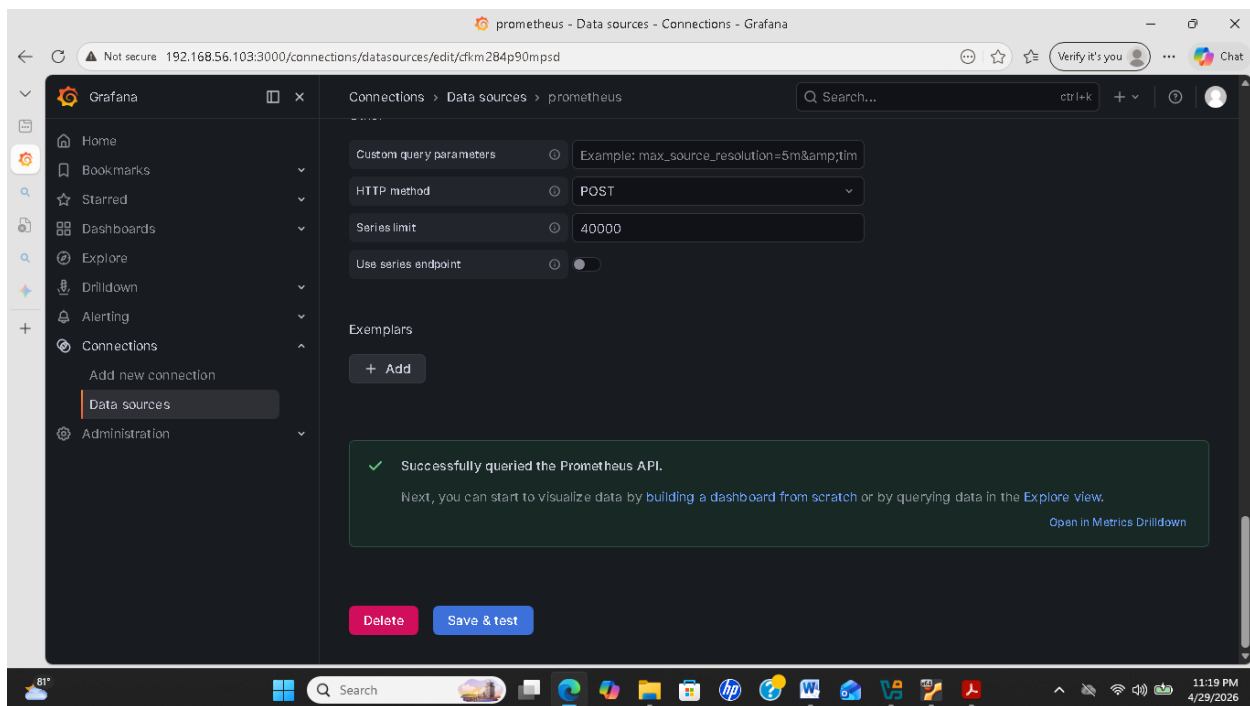
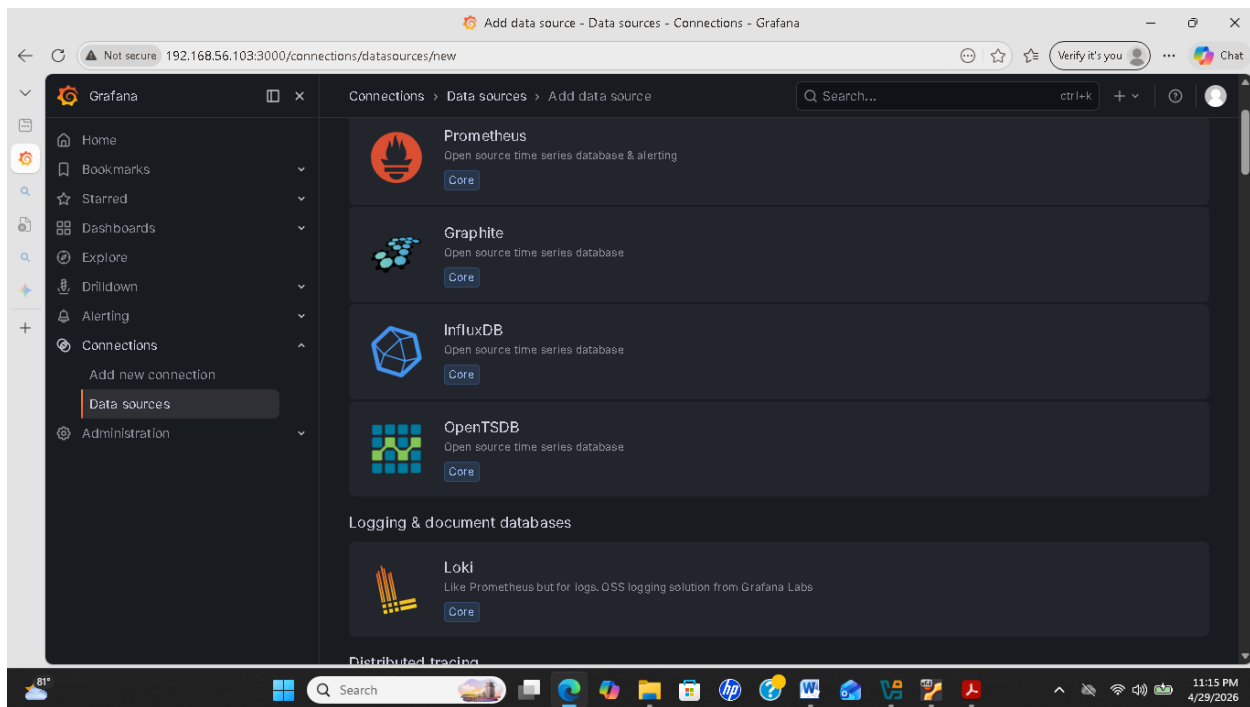
```
sudo systemctl daemon-reload
sudo systemctl enable grafana-server
sudo systemctl start grafana-serve
sudo systemctl status grafana-server
```

5. J'ai ajouté Prometheus comme source de données

URL : `http://localhost:9090`





Conclusion

La mise en place d'une supervision avec Prometheus et Grafana a permis d'améliorer la gestion et la stabilité de l'infrastructure de DevCloud.

Elle offre une surveillance en temps réel des ressources (CPU, RAM, disque, réseau et Nginx), facilitant la détection rapide des problèmes.

Cette solution garantit ainsi une meilleure disponibilité et performance du service web.